Algorithms and Flowcharts:

- A typical programming task can be divided into two phases:
- ***Problem solving phase***
  - produce an ordered sequence of steps that describe solution of problem.
  - this sequence of steps is called an ***algorithm.***
- ***Implementation phase***
  - implement the program in some programming language.

Steps in Problem Solving:

- First produce a general algorithm (one can use ***pseudocode***)
- Refine the algorithm successively to get step by step detailed ***algorithm*** that is very close to a computer language.
- ***Pseudocode*** is an artificial and informal language that helps programmers to develop algorithms. Pseudocode is very similar to everyday English.

Pseudocode & Algorithm:

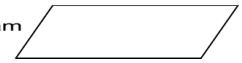
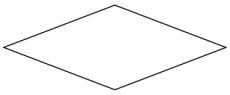
- **Example 1:** Write an algorithm to determine a student's final grade and indicate whether it is passing or failing. The final grade is calculated as the average of four marks.

The Flowchart:

A Flowchart

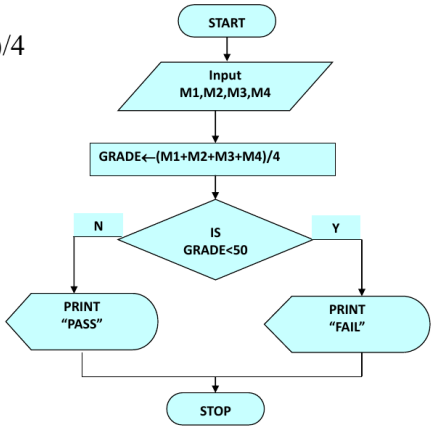
- shows logic of an algorithm
- emphasizes individual steps and their interconnections
- e.g. control flow from one action to the next

Flowchart Symbols:

Name	Symbol	Use in Flowchart
Oval		Denotes the beginning or end of the program
Parallelogram		Denotes an input operation
Rectangle		Denotes a process to be carried out e.g. addition, subtraction, division etc.
Diamond		Denotes a decision (or branch) to be made. The program should continue along one of two routes. (e.g. IF/THEN/ELSE)
Hybrid		Denotes an output operation
Flow line		Denotes the direction of logic flow in the program

Example 1:

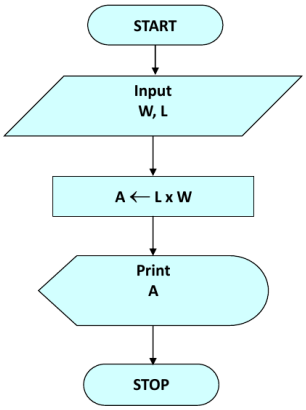
- Step 1: Input M1,M2,M3,M4
- Step 2: GRADE ← (M1+M2+M3+M4)/4
- Step 3: if (GRADE <50) then
  Print “FAIL”
  else
  Print “PASS”
  endif



Example 3:

Algorithm

- Step 1: Input W,L
- Step 2: A ← L x W
- Step 3: Print A



Example 4:

- Write an algorithm and draw a flowchart that will calculate the roots of a quadratic equation:

$$ax^2 + bx + c = 0$$

- Hint: **d** = sqrt ($b^2 - 4ac$), and the roots are:

  ***x*1** = $(-b + d)/2a$ and ***x*2** = $(-b - d)/2a$

Example 4:

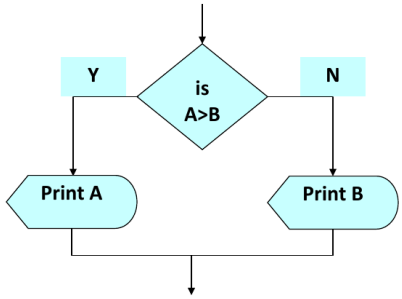
Pseudocode:

- *Input the coefficients (a, b, c) of the quadratic equation*
- *Calculate* **d**
- *Calculate* ***x*1**
- *Calculate* **x2**
- *Print* x1 *and* x2

IF–THEN–ELSE STRUCTURE:

- The algorithm for the flowchart is as follows:

If A>B then
print A
else
print B
endif



Relational Operators:

Relational Operators	
Operator	**Description**
>	Greater than
<	Less than
=	Equal to
≥	Greater than or equal to
≤	Less than or equal to
≠	Not equal to

Example 5:

- Write an algorithm that reads two values, determines the largest value and prints the largest value with an identifying message.

ALGORITHM

Step 1: *Input* VALUE1, VALUE2
Step 2: *if (*VALUE1 > VALUE2) *then*
MAX ← VALUE1
else
MAX ← VALUE2
endif
Step 3: *Print "The largest value is", MAX*

Example 6:

Step 1: *Input* N1, N2, N3
Step 2: *if* (N1>N2) *then*
if (N1>N3) *then*
MAX ← N1 [N1>N2, N1>N3]
else
MAX ← N3 [N3>N1>N2]
endif
else
if (N2>N3) *then*
MAX ← N2 [N2>N1, N2>N3]
else
MAX ← N3 [N3>N2>N1]
endif
endif
Step 3: *Print "The largest number is", MAX*

Example 6:

- **Flowchart: Draw the flowchart of the above Algorithm**

Example 7:

Write an algorithm and draw a flowchart to

a) read an employee name (NAME), overtime hours worked (OVERTIME), hours absent (ABSENT) and

b) determine the bonus payment (PAYMENT).